MGe
LABS

DayNight System v1.1.0

Thanks a lot for downloading this package!
If you enjoy it, consider leaving a rating on the store :)

Below you'll find instructions on how to get started.
All the C# code is documented and structured, providing a powerful API that is easily extensible.

If you have any questions, problems or suggestions, don't hesitate to reach out at: mgelabs.assets@gmail.com

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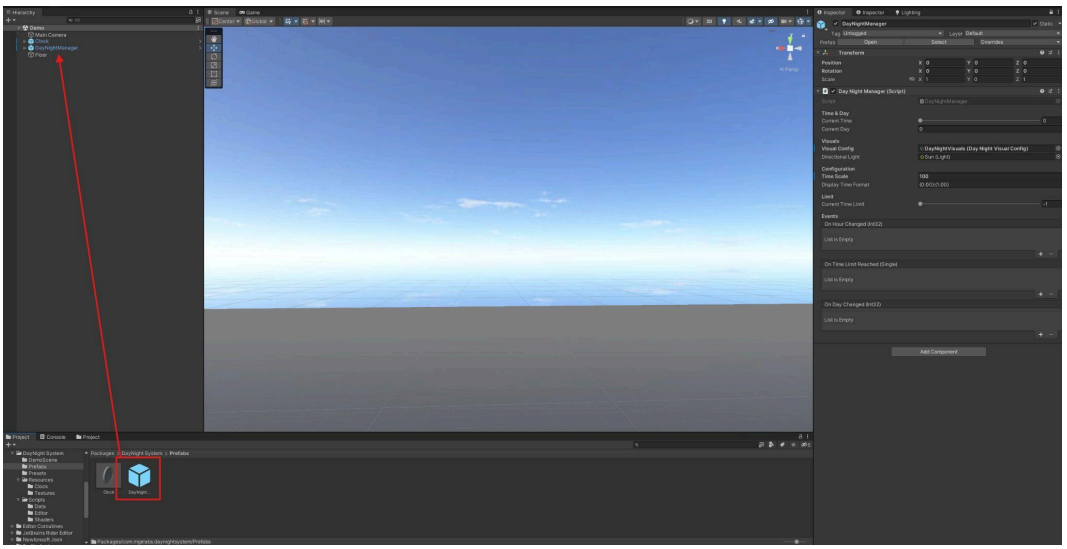
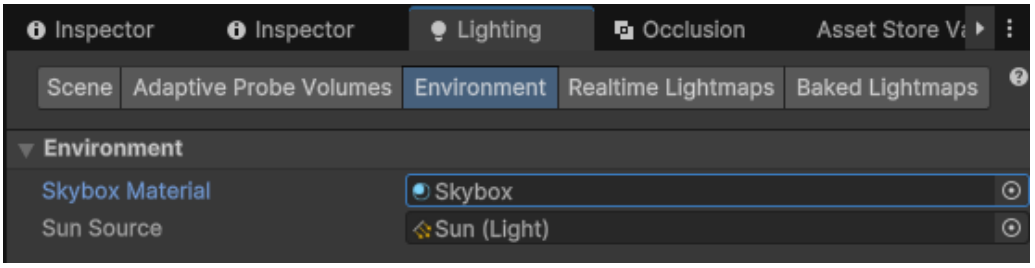
Quick Start

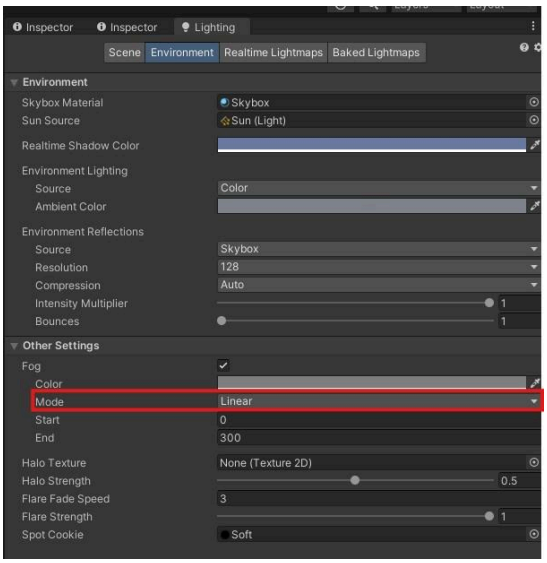
1) Drag the DayNightManager prefab into your scene:



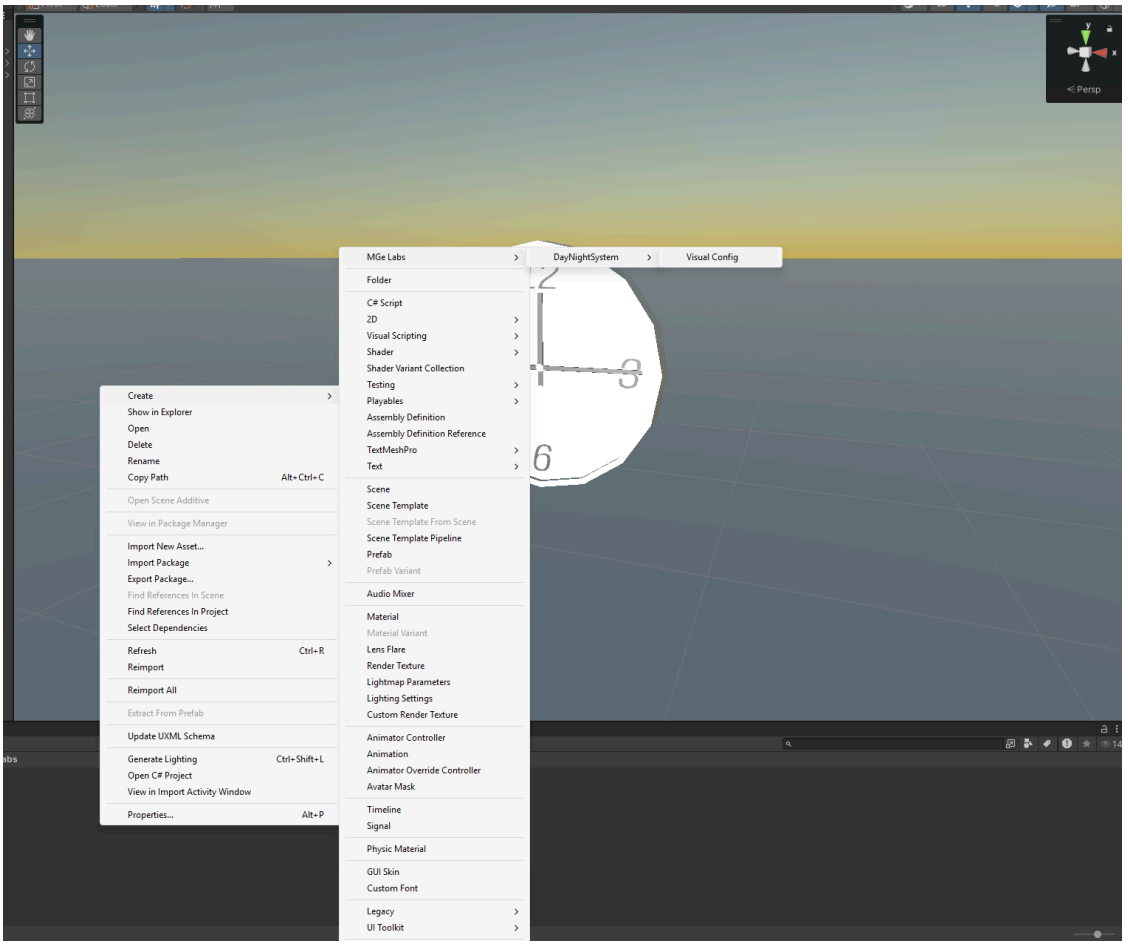
2) Set the Skybox material in the Unity Lighting tab to either your own Material that uses the DayNightSystem/TransitionSkybox shader, or the included Skybox material in the Resources folder:



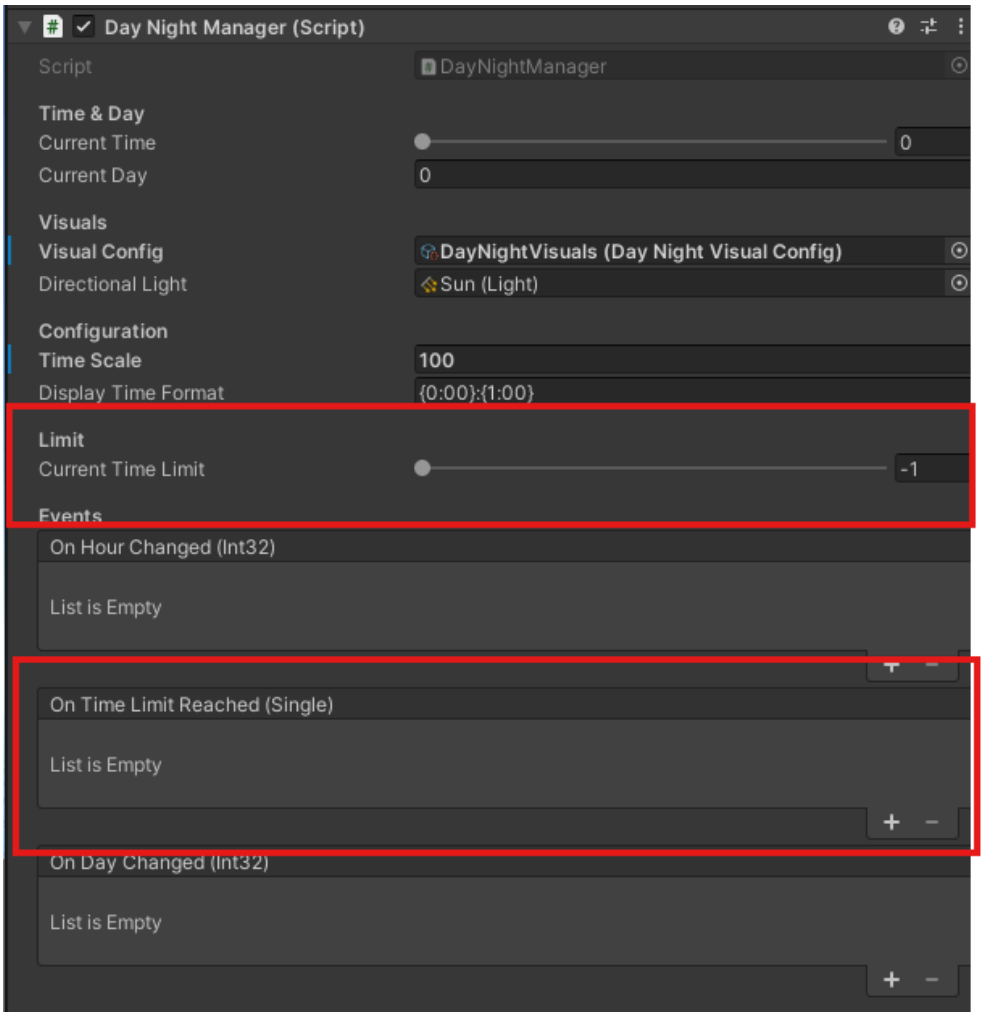
3) (Optional) Change fog mode to “Linear” if you want dynamic fog distances (in the Scene Lighting tab):



4) (Optional) Create your own visual preset:
You can add your own visual presets by creating the scriptable objects. You can find them in “Create/MGe Labs/DayNightSystem/Visual Config”. You’ll find detailed instructions below.
Make sure to assign your preset in the DayNightManager’s Visual Config field.
If you don’t want the visuals, leave the config field empty.

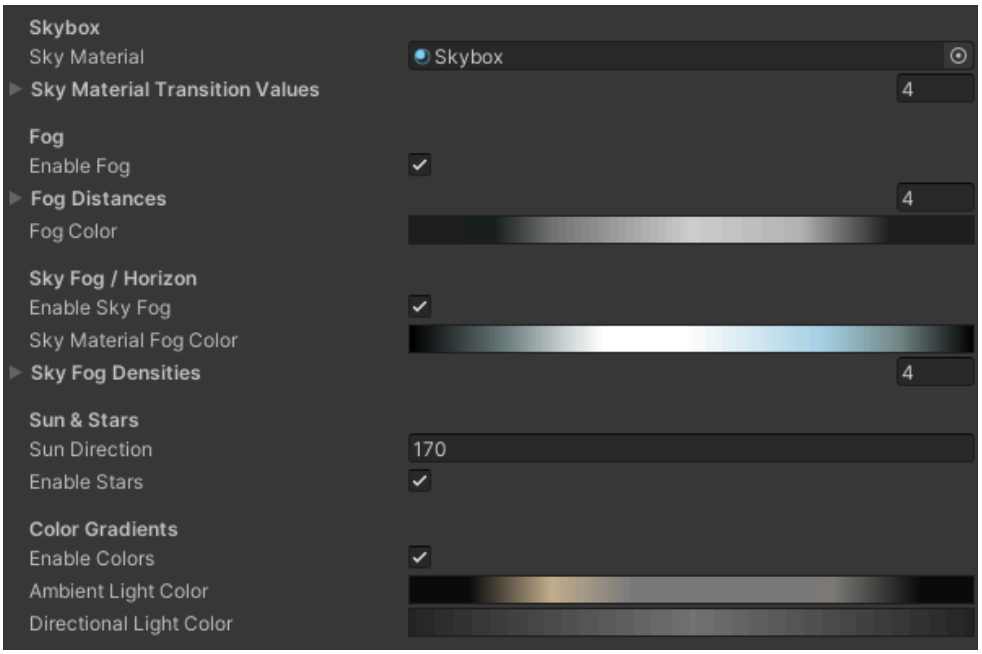


5) (Optional) Limit the Time
If you don't want a full day rotation, but instead have the time stop progressing at a certain time, you can set a limit. Once reached, the time won't progress further until you either call ResetTime() on the DayNightManager (which resets the time back to the initial start time), or by setting a sooner time than the limit and then calling ResetTimeLimit().
You can subscribe to the OnTimeLimitReached UnityEvent or OnTimeLimitReachedAction.



Visual & Shader Config

Visual Config



SkyMaterial: The Skybox material, you can use the existing one in the **Resources** folder, or create your own. For the skybox visuals to work, the material requires the DayNightSystem/TransitionSkybox shader.

Sky Material Transition Values: Controls how the skybox transitions visually over the course of the day. Values correspond to points in the day (0 = midnight/dark, 1 = noon/light).

Enable Fog: Enable dynamic fog distance & color

Sky Material Fog Color: The fog/horizon colors of the skybox. The gradient represents the transition colors over the cycle.

Sky Fog Densities: The sky fog densities over the cycle (should be 4).

Sun Direction: The direction of the sun.

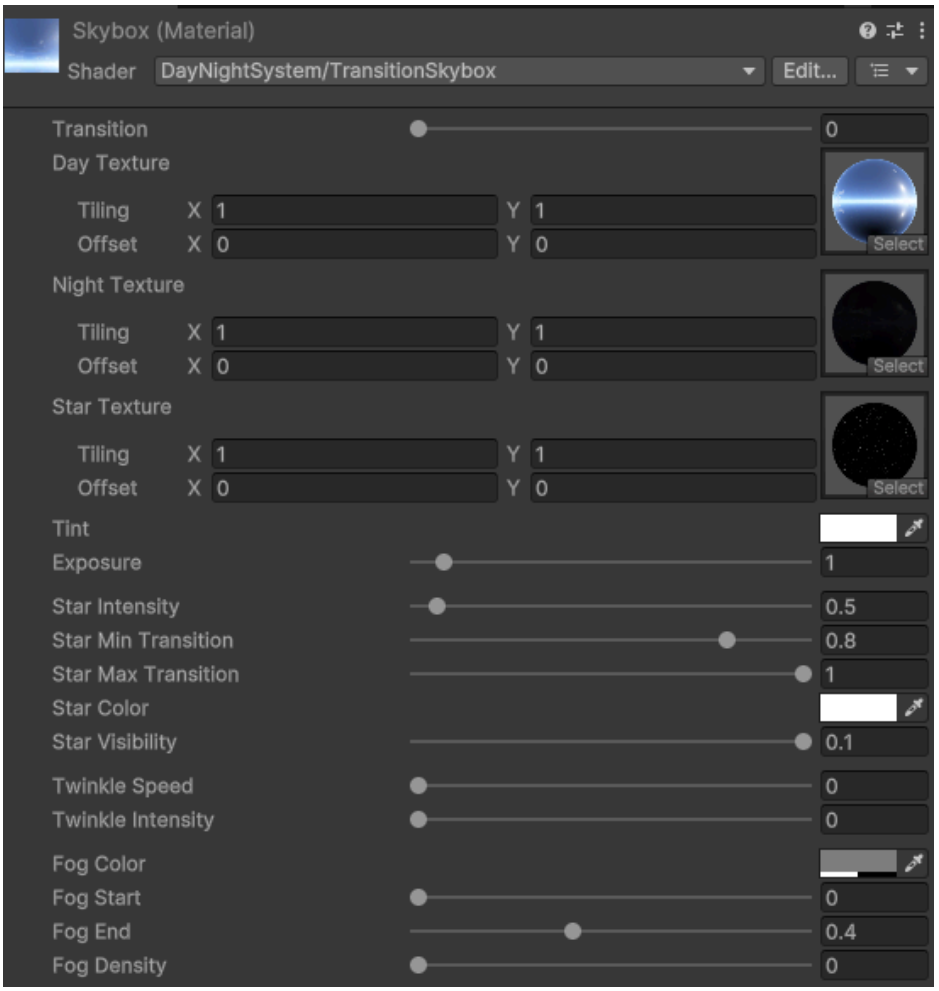
Enable Stars: Won't do much by itself; if turned off, it will disable stars in the skybox material, if on, it keeps your stars config.

Enable Colors: Enable ambient and light color changes

Ambient Light Color: The Unity Environment Light "Ambient Color" over a cycle.

Directional Light Color: The directional light / sun color over a cycle.

Skybox Shader



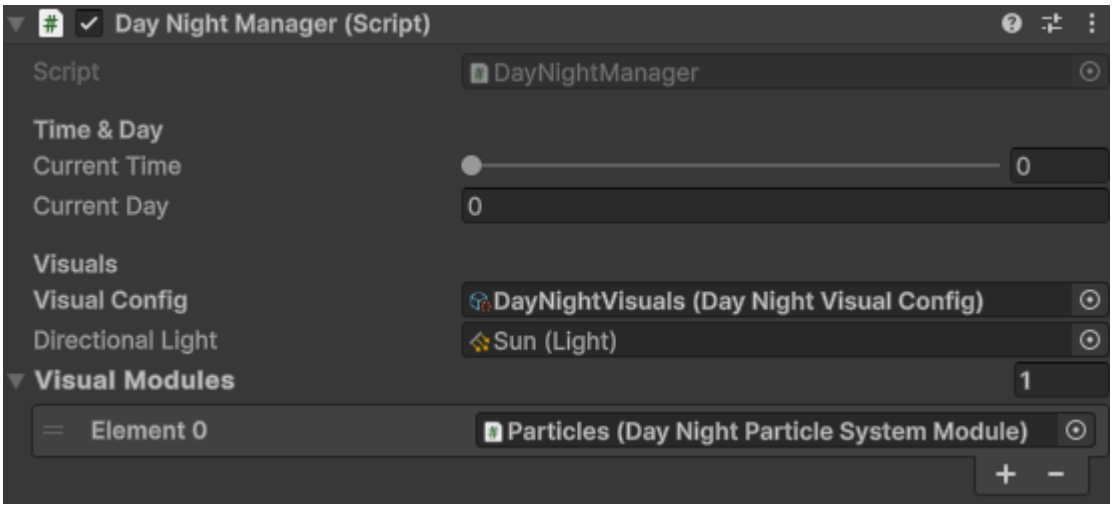
Transition: The transition amount between day and night (0-1)
Day Texture: The cubemap texture of the day sky
Night Texture: The cubemap texture of the night sky
Star Texture: The cubemap texture of the stars
Tint: Change the overall color
Exposure: Adjusts the brightness of the sky

Star Intensity: Controls the brightness of the stars (0-10)
Star Min Transition: The transition value where stars begin to appear
Star Max Transition: The transition value where stars are fully visible
Star Color: Tint color applied to the stars
Star Visibility: Controls the mix between the stars and the sky
Twinkle Speed: Speed of the star twinkle animation
Twinkle Intensity: Strength of the twinkle effect

Fog Color (Sky): The fog color applied to the skybox
Fog Start: The normalized height where fog begins to appear
Fog End: The normalized height where fog is at full strength
Fog Density: Controls how quickly the fog blends with the sky

Visual Modules

You can create modules that you can plug into the DayNightManager, which then react to time changes in order to represent a visual state.
There is a **DayNightVisualModule** base-class, that you'll have to extend.

There is a **DayNightParticleSystemModule** MonoBehaviour as an example, you can assign particle systems to it, and it will change the particle's colors, lifetimes and emission values based on the current daytime.
This can be useful for dust, fog,.. particles, which will be less visible at night.

In order for the modules to work, you have to assign them to the visual modules in the **DayNightManager**: